

Assignment-I(A)

Function Spaces

(For Submission)

1. Let (X, ρ) and (Y, σ) be two metric spaces and $f : X \rightarrow Y$ be a function. Show that: f is uniformly continuous if and only if for any two sequences $\{u_n\}$ and $\{v_n\}$ in X , if $\lim_{n \rightarrow \infty} \rho(u_n, v_n) = 0$ then $\lim_{n \rightarrow \infty} \sigma(f(u_n), f(v_n)) = 0$.
2. Provide an example of a descending sequence of closed non-empty subsets of $\mathbb{R}$ whose intersection is empty.
3. Consider the metric space $(C[0, 1], \rho_\infty)$, where ρ_∞ is the usual sup-metric.

(i) Let

$$D := \{f \in C[0, 1] : f \text{ is differentiable on } (0, 1)\}.$$

Is D complete? (Justify)

(ii) Let

$$\mathcal{L} := \{f \in C[0, 1] : f \text{ is a Lipschitz function}\}.$$

Is $\mathcal{L}$ complete? (Justify)

4. Let K be a compact subspace of a metric space (X, ρ) . If $y \in X \setminus K$, prove that there exists a point $x \in K$ such that

$$\rho(x, y) = \text{dist}(y, K).$$

Give an example to show that the conclusion may fail if ‘compact’ is replaced by ‘closed’.

5. Let A and B be two subsets of a metric space (X, ρ) . Define

$$\text{dist}(A, B) = \inf\{\rho(a, b) : a \in A, b \in B\}.$$

If A is compact and B is closed, show that $A \cap B = \emptyset$ if and only if $\text{dist}(A, B) > 0$.

6. Let K be a compact subset of a metric space X and V be an open set containing K . Show that there is an open set U for which $K \subseteq U \subseteq \overline{U} \subseteq V$.

7. Let (X, ρ) be a compact metric space and $T : X \rightarrow X$ such that

$$\rho(T(x), T(y)) < \rho(x, y) \quad (\forall x, y \in X, x \neq y).$$

Prove that T has a unique fixed point.

8. True or False: Let $\{f_n\}$ be a sequence of functions on $X \subseteq \mathbb{R}$. If $\{f_n\}$ is uniformly convergent on every subset of X that is closed in $\mathbb{R}$ then $\{f_n\}$ is uniformly convergent on X . (Justify)

9. Justify the following True or False questions:

(i) Let $\{f_n\}$ be a sequence of continuous functions on $\mathbb{R}$ which converges uniformly to a function f on $\mathbb{R}$ and g be a continuous function on $\mathbb{R}$. Then $\{g \circ f_n\}$ converges uniformly to $g \circ f$.

(ii) Let $\{f_n\}$ be a sequence of continuous functions on $\mathbb{R}$ which converges uniformly to a function f on $\mathbb{R}$ and g be a continuous function on $\mathbb{R}$. Then $\{f_n \circ g\}$ converges uniformly to $f \circ g$.

Assignment-I(B)

Function Spaces

(Some Additional Practice Problems)

1. In a metric space X , show that a Cauchy sequence converges if and only if it has a convergent subsequence.
2. Let K be a compact subset of a metric space X and $x \in X \setminus K$. Show that there exists an open set U containing K and an open set V containing x for which $U \cap V = \emptyset$.
3. Let (X_1, ρ_1) and (X_2, ρ_2) be two metric spaces. Define, $\rho_1 \times \rho_2 : X_1 \times X_2 \rightarrow \mathbb{R}$ by

$$\rho_1 \times \rho_2((x_1, x_2), (y_1, y_2)) = [(\rho_1(x_1, y_1))^2 + (\rho_2(x_2, y_2))^2]^{1/2}.$$

Prove that

- (i) $\rho_1 \times \rho_2$ is a metric on $X_1 \times X_2$.
 - (ii) If X_1 and X_2 are compact then $X_1 \times X_2$ is also compact.
 - (iii) If X_1 and X_2 are complete then $X_1 \times X_2$ is also complete.
4. A metric space X is said to be locally compact if every point x has a compact neighbourhood, that is, for each point x , there is a compact set K_x such that $x \in K_x^\circ$ (interior of K_x). Prove that
 - (i) $\mathbb{R}^n$ is locally compact.
 - (ii) The set of rationals $\mathcal{Q}$ is not locally compact.
 5.
 - (i) Determine whether boundedness is preserved by uniformly continuous functions between metric spaces.
 - (ii) Determine whether totally boundedness is preserved by uniformly continuous functions between metric spaces.
 - (iii) Determine whether a function is uniformly continuous if it preserves total boundedness.
 6. Let (X, ρ) be a complete metric space and $T : X \rightarrow X$ be a map. Suppose, $\exists$ $0 < K < 1/2$ such that

$$\rho(T(x), T(y)) \leq K[\rho(x, T(x)) + \rho(y, T(y))] \quad (\forall x, y \in X).$$

Prove that T has a unique fixed point.

Hints: Follow the construction in the proof of the Banach Fixed Point Theorem, which we have seen in class.